

ADI BANERJEE

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EDUCATION

University of Arizona

Bachelor of Science, Computer Science | Minor: Data Science and Statistics

2019-2023

GPA: 4.0

- Upper Division Coursework: Algorithms, Databases, Automata, Compilers, Linear Models, Machine Learning, Computer Vision
- Academic Awards: Summa Cum Laude, 4x Highest Academic Distinction, \$120000 Merit Scholarship
- Research Awards: Excellence In Undergrad Research, 2x Galileo Circle Scholar

PUBLICATIONS

Published, University of Arizona

- Pyarelal, A., **Banerjee, A.**, Barnard, K. (2022). Modular Procedural Generation for Voxel Maps. In: Gurney, N., Sukthankar, G. (eds) Computational Theory of Mind for Human Machine Teams. AAAIFSS 2021. Lecture Notes in Computer Science, vol 13775. https://doi.org/10.1007/978-3-031-21671-8_6.

Ongoing Research, University of Arizona

- **Transformer-based OCR for Mathematical Expressions:** Developing a transformer-based OCR model to read mathematical expressions from images and convert them into grammatically correct LaTeX. The model is trained using a hybrid two-stage learning approach: transfer learning followed by reinforcement learning. A key challenge in building this model is ensuring grammatically correct LaTeX output using an ad-hoc abstract syntax tree to represent loss for a context-sensitive, Turing-complete language like LaTeX, which lacks a standard Backus-Naur Form (BNF).
- **Vector Quantized Reinforcement Learning:** Applying vector quantization (VQ) to reinforcement learning. The project involves developing a framework for learning prototype vectors that a model can use in its learned feature space. Our initial experiments have yielded promising results, particularly with a new cosine similarity loss-based approach to learning the prototype vectors. We have a paper in the draft stage. Preprint: <https://arxiv.org/pdf/2211.06733>

PRESENTATIONS

Galileo Circle Awards, University of Arizona

2021

- Presented a procedural generator I designed and implemented for the ToMCAT (Theory of Mind-based Cognitive Architecture for Teams) highlighting the reasons for using a flexible axis-aligned bounding box approach to modularization, which allows for more intuitive customizations during map generation.

EXPERIENCE

Amazon, Software Development Engineer 2

July 2025 - Present

- Developed agentic capabilities that let dropship vendors diagnose delivery performance drops and warehouse suspensions through Amazon's vendor-facing AI assistant, via an MCP server and skills backed by DuckDB over S3 Parquet, built spec-first with OpenSpec, evaluated with Langfuse traces and LLM judges, and authorized per user by task-based role.
- Proposed and developed a documentation-drift detector that models a wiki corpus as a Neptune graph and uses a Rust Lambda on Bedrock to find stale documentation and generate reviewable edits, with a self-developed topic-based diff algorithm that feeds the model 200 tokens of graph topics per query instead of 50K tokens of page content, analyzing 30+ pages in under 5s from a 48ms cold start. Won OSS 2026 and Spark 2026, two years in a row after 2025 wins at both for co-developing a LangGraph multi-agent tech-debt profiler which auto cleaned up deprecated code and cloud resources.
- Designed the ingestion layer and co-developed a sourcing-explainability agent that streams buying-decision changes from 3 upstream systems through Kinesis and Lambda into OpenSearch (30M documents), giving the agent real-time cross-system access in one 100ms lookup in place of multi-hop API resolution, with a 7.5s p95 time to first token against a 20s requirement and 85s end-to-end answers from a 198s p95 baseline.
- Co-designed and led implementation and launch of embedded QuickSight dashboards in Vendor Central across 5 marketplaces (React, Java, CDK), retiring a 7-service legacy analytics stack so analysts publish a new metric without engineering work.

Amazon, Software Development Engineer

January 2024 - July 2025

- Led suspension system overhaul with EventBridge batch processing and DynamoDB distributed rate limiting, improving execution from 2 days to 5 min and throughput from 1 to 200 requests/min.
- Delivered 4 minor and 2 major projects over 5 months, leading to SDE-2 promotion recommendation 5 months after start.
- Presented network optimization data analysis and designed workflow affecting 230MM+ configs as tech lead across 3 teams.

- Migrated 26 vendor email use cases across 12 services off a 10-year-old Tier-1 notification service to SES and an internal platform, with 11 service-level feature flags on a graduated rollout and a 0% error rate since launch.

University of Arizona, Research Engineer

July 2023 - January 2024

- Designed deep RL transformer math OCR models, achieving a 6% increase over SOTA using PyTorch, and Stable Baseline.
- Overhauled MathML parser with Nom in Rust improving traceability for 30% of tag syntax errors during AST generation.

University of Arizona, Undergraduate Research Assistant

September 2019 - May 2023

- Designed modular procedural generation framework in C++ for emulating RL environments, and 2nd authored AAAI paper.
- Sole recipient of the Research Excellence Award for Computer Science and 1 of 4 recipients of the Galileo Circle Award, twice.

Amazon, Software Development Engineer Intern

May 2022 - August 2022

- Designed a React in TypeScript full-stack internal tool and deployed to CloudFront with proxy-based authentication for securely monitoring millions of purchase orders globally, enhancing operational transparency across 2 Amazon orgs.
- Automated non-relational to relational data transformation pipeline with AWS CDK, DynamoDB, Kinesis Datastream, Kinesis Firehose, Lambda, S3, Glue, Athena, and IAM, achieving a 98% improvement in data availability.

Tech Core, Lead Software Developer

February 2021 - May 2022

- Led web and AR/VR projects using NodeJS and .NET for 10+ clients across 6+ university departments and startups.
- Optimized SQL queries and JS through code profiling, improving web app load times by 80.51% across 3 services.

AWARDS AND SCHOLARSHIPS

Galileo Circle | 2021-2023

- \$1000 scholarships awarded to undergraduate and graduate students who demonstrate exceptional potential in the physical, mathematical, environmental, cognitive, or life sciences. 2-time awardee.

Academic Merit Scholarship | 2019-2023

- A competitive undergraduate tuition scholarship offered annually to a select number of international students attending Arizona. \$120,000 received in tuition scholarship.

Summa Cum Laude | 2023

- Awarded to students graduating with a GPA of 4.0 over 4 years.

Highest Academic Distinction | 2019-2023

- 4-time recipient from 2019-2023. Awarded to students maintaining a 4.0 GPA for an academic year.

Excellence in Undergraduate Research | 2023

- Awarded to one student in a department for significant contribution, originality, creativity, and independence in research.

Dean's List With Distinction | 2019-2023

- 8-time recipient from 2019-2023. Awarded to students maintaining a 4.0 GPA for an academic year.

PROJECTS

Whisper | *Java 21, Rust (JNI), Spring Boot, React, Docker, Helm*

- Built a shared daily safe-word service with Google OAuth, timezone-aware rotation scheduling, JNI-based Rust word generation, HikariCP connection pooling, Bucket4j rate limiting, and Prometheus metrics via Micrometer.
- Packaged as a 4-stage Dockerfile and Helm chart (Deployment, Service, CronJob, Ingress) for Kubernetes deployment.

Hindsight | *Python, FastAPI, React, Vercel*

- Built a keyless backtest visualizer for equity trading strategies with real-time portfolio charting, trade annotation, and drawdown analysis. Deployed on Vercel with GitHub Actions CI.

dups | *Go, Goroutines, SHA-256*

- Built a concurrent duplicate file finder with a two-pass hashing strategy (4KB partial hash for files over 1MB, full SHA-256 for confirmed candidates), channel-based worker pool, interactive per-group deletion, and cross-platform builds via goreleaser.

Host Management System | *Python, Apache, AWS Boto*

- Architected an EC2 host management system leveraging cryptographic techniques for secure user authentication and session management. Implemented features encompassing host monitoring, dynamic fleet scaling, and controlled termination, employing encryption protocols to safeguard against potential man-in-the-middle attacks.

TaylorGPT | *PyTorch, Lightning, Numpy*

- Designed a 1.4 million parameter character-level transformer decoder model to Taylor Swift lyrics from scratch using PyTorch including custom implementations for masking, multi-head attention, and transformer blocks along with model profiling, logging, and multi-GPU training.

Tau Compiler | *Python, PyTest*

- Developed an LL1 Recursive Descent compiler for Tau, comprising of a lexer, parser, decorator, and generator including syntax analysis, scoping, type checking, optimization, and memory management.

Cat(G)AN | *Python, TensorFlow, Keras, Matplotlib*

- Implemented a Generative Adversarial Network (GAN) with 3.5 million parameters from scratch in NumPy, TensorFlow, and Keras to generate cat images.

Start Journey | *Python, PyTorch, Diffusers, FastAPI, React, Vite*

- Built a text-to-image web app that runs Stable Diffusion behind a FastAPI backend serving a React frontend, streaming generation progress live and running on CUDA, CPU or Apple MPS from one install.

MEOWDEL | *PyTorch, Lightning*

- Built a DCGAN from scratch that generates 64x64 cat faces, comparing a baseline against spectral normalization and R1 penalty, with a writeup of a mode-collapse fix.

PyGanttter | *Python, Plotly*

- Built a typed CLI that renders Gantt charts from CSV with dependencies, critical path, milestones and progress, as interactive HTML or static PNG.

ResumeGPT | *Python, PyTorch, Streamlit, ChromaDB*

- Designed and deployed a Streamlit application with LangChain, ChatOpenAI, and ChromaDB that uses LLMs to answer questions about me including customized underlying vector DB logic.

Signature | *Python, OpenCV, Pillow, Matplotlib*

- Designed a multi-platform desktop application that lets you create an e-signature from a photograph with custom implementations for image recognition logic that combined normalization and thresholding approaches to extract signatures.

AI-PlaysSnake | *Python, NEAT, PyGame*

- Implemented the game Snake and a genetic neuro-evolving reinforcement learning model to beat it including implementations for the underlying physics engine and graphics for the game.

OUTREACH, SOCIETIES AND EXTRACURRICULARS

Amazon SCOT Spark Hackathon, Evaluation & Verification Award 2026

- Merged the documentation-drift detector into a knowledge-base governance product with a team of 8; proposed the integration points, guided scope on graph storage, Rust, grounding and cross-workspace extensibility, and ran the final dry run. Awarded among 100+ teams and 29 finalists.

Amazon SCOT OSS AI Hackathon, Runner-up, Technical Excellence 2026

- Proposed and developed a documentation-drift detector (Rust Lambda, Bedrock, Neptune graph, topic-based diff algorithm) that finds stale wiki pages and generates reviewable edits; 18,800+ lines across 3 packages shipped to beta in 2.5 days.

Amazon SCOT Spark Hackathon, Winner 2025

- Developed a multi-agent code analysis agent in collaboration with a team of 5 to automatically find and build dependency graphs for deprecated code, which resulted in two hackathon wins with 1000+ Amazon teams participating globally.

Google Developers Student Club, Chapter Founder and President 2020-2021

- Led a team of 6 to organize 30+ workshops in Web Dev, Machine Learning, and Career Building for 180+ members.
- Partnered with 50+ GDSC chapters worldwide fostering innovation and increasing overall CS club engagement by 132%.

MLH: Hack Georgia Tech, Mentor 2020

- Invited to mentor hacker teams at Hack Georgia Tech Hackathon with 2500+ attendees.

MLH: Hack Arizona, Winner 2020

- Received Amazon's Most Impactful Use of Data-Driven Technology at Hack Arizona among 200+ contestants and 50+ teams for developing a machine learning model for detecting distracted drivers with AWS Sagemaker.

MLH: To the Moon and Hack, Winner 2020

- Received In-Flight Anomaly prize among 330+ contestants for Project Space Station: a spin-off Space Invaders, and an AI trained via neuroevolving genetic reinforcement learning methods to beat it.

University of Arizona Residence Hall Association, Council Member

2019-2020

- Promoted sustainability practices and awareness across dorms and campus related to various social, environmental, and medical issues for which I received two RHA Person of the Week nominations.
- Organized and voted on various dorm and RHA programs, events, budgets, and issues.

Blue Chip Leadership, Volunteer

2019-2020

- Participated in community service projects with local AIDS organizations like SAAF (Southern Arizona Aids Foundation) and sustainability organizations like Tucson Trees Please.

Google Developer Challenge, Winner

2019

- Received 2nd place award for solving algorithm problems at Google Tech Challenge among 50+ contestants.

SKILLS

Languages: Java, Python, Rust, Go, TypeScript, C++, SQL

AI & Agents: MCP, Bedrock, LangGraph, Strands, Langfuse, LLM-as-judge evals, Neptune, OpenSpec, Kiro, Claude Code, PyTorch, Diffusers

Frameworks & Infrastructure: Spring Boot, FastAPI, React, AWS CDK, Lambda, DynamoDB, OpenSearch, SQS, Kinesis, EventBridge, S3, Parquet, DuckDB, Redshift, QuickSight, Docker, Kubernetes

ADDITIONAL LINKS

- Personal Website: <https://adi-ua.github.io/>
- Publications and In Progress Research: <https://adi-ua.github.io/research/>
- LinkedIn: <https://www.linkedin.com/in/adi-ua/>
- GitHub for all projects: <https://github.com/Adi-UA>